UN Open GIS Monthly VTC

20 May 2026

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Agenda

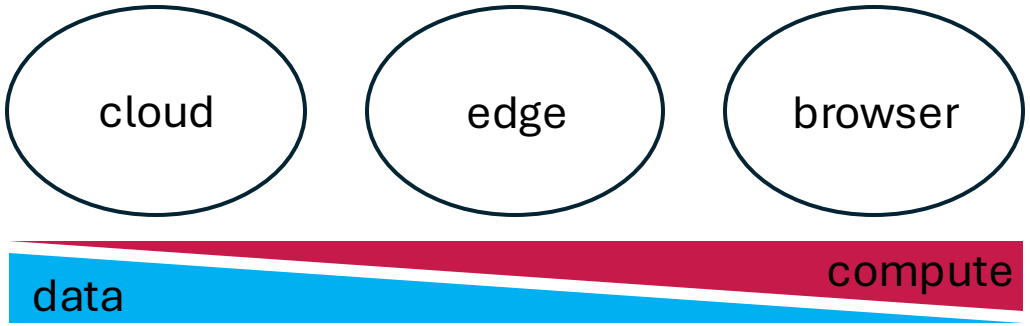
1. Comments by co-chairs, introduction of new participants
2. DWG and pilot project reports – ascending order
    1. DWG1 & pilot projects: Hybrid GIS Infrastructure – Timur
    2. DWG2: Capacity Building – Diego
    3. DWG5: Geo-AI – Zhongxin
    4. DWG6: OpenDroneMap – Stephen
    5. DWG7: Smart Maps – Hidenori
    6. DWG8: IGIF implementation – Ariel
3. Any Other Business
    1. Domain name renewal - unopengis.org

DWG-7: Smart Maps

Keep web maps open. Test new technologies. Practice software freedom.

From Bulk Data Sharing to Cloud-Native Smart Maps

- **Field reality**
  - UAV imagery needs fast access and sharing. (Sierra Leone)
  - Geospatial data locked in ZIP files. (Hokkaido)
- **Current limitation**
  - Bulk download or slow TMS. No instant partial access.
- **Cloud-Native approach**
  - Separate processing and storage
  - Move processing close to data
  - Use data by reference (don’t copy)

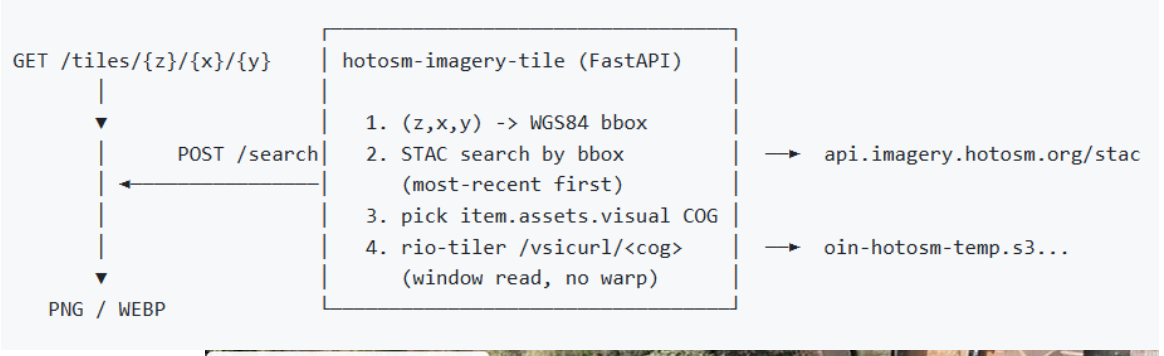


PoC: On-demand imagery access from STAC

(No Bulk Download) - gh:yuiseki/poc-cng-hotosm-imagery-tile

STAC (metadata) → GeoTIFF (cloud storage) → On-demand tile generation (Function as a Service / edge) → Web Map

Tiles generated when requested.



[Value]

1. Instant access via URL + STAC
2. No bulk download required
3. Scalable processing (cloud→edge)

[Next step]

- Compare with static hosting.
- Feeds into operational deployment.



Five-Month Plan: From PoC to operational stack

Build on the PoC to deliver an operational cloud-native pipeline.

1. **Partnership established**
   Sierra Leone / Hokkaido use cases defined.
2. **End-to-end pipeline validated**
   Re-deploy cloud-native data to edge and portable environments
   Evaluate end-to-end pipeline from dataset – edge node – browser client.
3. **Operational stack delivered**
   High-resolution imagery accessible on demand (no bulk download required).

AOB: Domain Name Renewal - **unopengis.org**

- The domain unopengis.org was due to expire in May 2026.
- Prof. Ki-Joune Li has generously supported the UN Open GIS Initiative by securing an additional five-year domain registration.
- We would like to express our sincere appreciation for his generous contribution and continued commitment.